

is a router through which a client connection request is sent either through a hop or directly to the database server.

Oracle Connection Manager

- A router through which client connection request may be sent either through a hop or directly to the database server.
- ⇒ The Oracle Connection Manager is a part of Oracle, serves as a router to facilitate communication between databases that use different network protocols.
- ⇒ It also supports multiplexing and access control.

Key features:

Protocol compatibility:

- It enables communication between databases using different network protocols.
- Allowing data transfer regardless of the protocols used on remote servers.

Supported protocols:

- It supports various protocols such as IPC, Named pipes, SDP, TCP/IP with SSL.

Components of OC4J: CMADMIN, CMGW, CMGW, CMADMIN

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Routing and load balancing:
 ⇒ It offers multiple access paths to handle different client requests. The connection manager selects the most suitable path based on availability and network load.

Firewall functionality:

- In intranet environment, it can function as a firewall for Oracle net traffic. Filtering rules can be set based on criteria like host names or IP addresses, services names etc.

Enhanced security:

- It enhances firewall security by filtering client access based on predefined rules.

Connection Manager Process

1. CMGW process (Connection Manager Gateway):

- Registers with CMADMIN and

the connection manager manages the connection to support connections within homogeneous networks, reducing the number of physical connections maintained by the database.

Two main processes and a control utility are associated with connection manager.

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⇒ The listener process listens for incoming connection requests. By default this process listens on port 1521. It acts as a hub for the clients to listeners and relay. It manages data between them.

2. CMADMIN (Connection manager administrative):

⇒ A multithreaded process responsible for all administrative tasks and issues.

⇒ Handle administrative tasks such as CMGW registration, maintaining listener information, updating service cache and managing source route information.

⇒ The CMADMIN process is tasked with identifying all listener processes that support at least one database.

⇒ Using Oracle Internet Directory, CMADMIN performs the following tasks:

- Locates local services
- monitors registered services
- maintains client address info

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Configuring the Oracle connection manager:

⇒ The cman.ora file contains the configuration parameters for the Oracle connection manager.

⇒ The file contains protocol addresses of the listening gateway process, access control parameters and control parameters.

The connection manager control utility (CMCTL):

⇒ The cmctl provides administrative access to CMADMIN and CMGW.

⇒ The connection manager is started via cmctl command.

Syntax:

⇒ cmctl command process type

⇒ The commands are broken into four basic types:

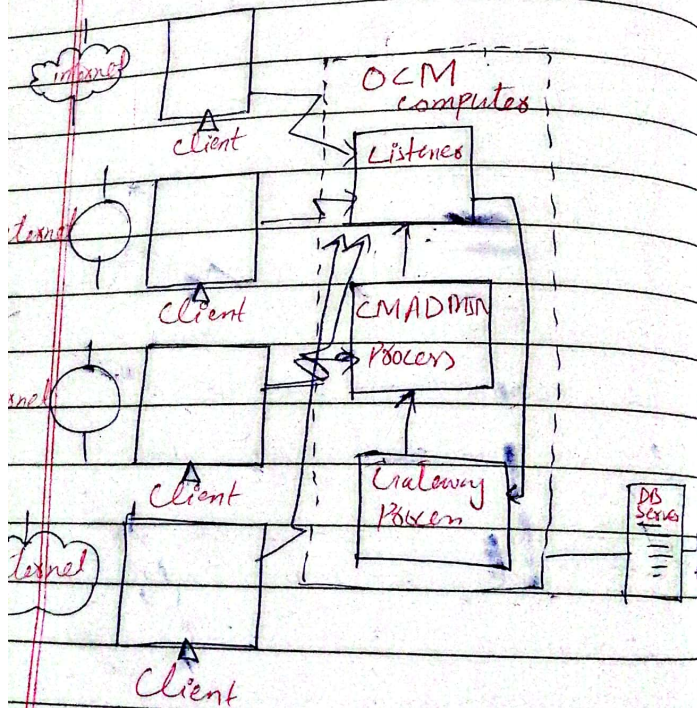
- Operational commands such as start.
- Modifier commands such as set.

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- o Informational commands such as show.
- o Command utility operations such as exit.

Architecture of OCM:



A connection is a communication pathway b/w a user process and oracle db instance. established → available for communication mechanism.

Topic: Oracle net Connections

⇒ Oracle net connections enable clients to establish and maintain network sessions with oracle database servers. This involve setting up the communication pathway and ensuring secure data exchange.

Key Components

Client Configuration:

tnsnames.ora: A file that maps service names to network addresses.

sqlnet.ora:

Contains global configuration settings for oracle net services

Server Configuration:

listener.ora: Defines the network protocols and addresses on which the oracle net listener will listen for connection request.

3. Connection process:

- o Service Naming
- o Listens interactions
- o Data transformation

Types of connection:

1. Dedicated server connections:

Each client connection is handled by a dedicated server process.

Suitable for heavy or long-running database operations.

2. Shared server connections:

multiple client connections share a pool of server processes, optimizing resource usage and scalability.

Security features:

- ⇒ Encryption
- ⇒ integrity check
- ⇒ Authentication